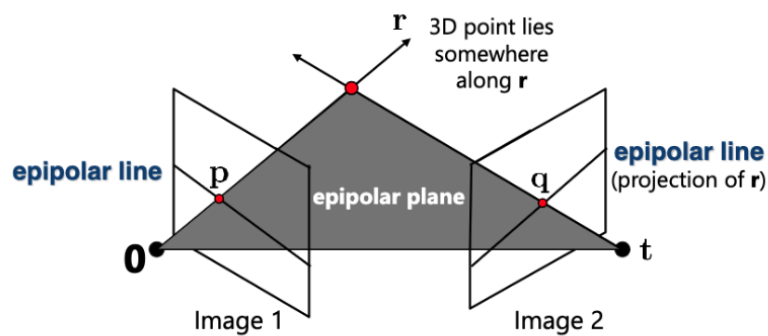


stereo: epipolar geometry



What Else Uses a Specific Number of Points?

Different geometric transformations preserve different properties (like parallelism, angles, or lengths) and have varying degrees of freedom. Here is how homography compares to other common transformations:

Transformation	Degrees of Freedom (Unknowns)	Minimum Points Needed	What it Preserves
Translation	2 (\$x\$, \$y\$ shifts)	1 point	Orientation, size, angles, parallelism
Euclidean (Rigid)	3 (2 shifts + 1 rotation)	2 points (Overdetermined)	Size, angles, parallelism
Similarity	4 (Rigid + uniform scaling)	2 points	Angles, shapes, parallelism

Affine	6 (Linear warp + translation)	3 points	Parallelism, ratio of areas
Homography (Projective)	8 (Full perspective warp)	4 points	Straight lines

Other Key Multi-View Geometry Point Counts

Beyond single-plane transformations, setting up relative 3D camera geometry requires specific point minimums:

- Essential Matrix ($\mathbf{E}$): Needs 5 Points. Used when intrinsic camera parameters are known to find the relative rotation and translation between two calibrated cameras.
- Fundamental Matrix ($\mathbf{F}$): Needs 7 or 8 Points. Used when cameras are completely uncalibrated. The linear configuration (8-point algorithm) uses SVD to solve for the epipolar geometry between two views.
- Perspective-n-Point (PnP): Needs 3 Points (P3P). If you have a known 3D model, you need a minimum of 3 world-to-image point matches to calculate the exact 3D position and orientation (pose) of the camera.

Diffusion Model Theory

- Forward process adds small Gaussian noise at each step
 - Markov chain assumption enables step-by-step reconstruction
 - Small steps preserve path information vs large noise jumps
 - No analytical method for determining optimal step count
 - Heuristics used with stopping criteria due to computational expense
- Joint distribution breakdown using chain rule
 - $q(x_1 \text{ to } x_T) = \text{product of conditional distributions}$
 - Each step modeled as normal distribution with physics-based parameters
 - Final step reaches standard Gaussian white noise

Reverse Process & Learning

- Goal: learn reverse process $p(x_{t-1} | x_t)$
- Modeled as Gaussian with neural network learning mean μ_θ and covariance σ_θ
- DDPM paper: σ_θ set as identity matrix with scaling parameter
- Alternative representation using noise parameter ϵ instead of full signal estimation
- Extremely computationally expensive process

DDRM Enhancement (Denoising Diffusion Restoration Models)

- Addresses known degradation: $y = Hx + n$ (blurring operator H)
- Uses Singular Value Decomposition (SVD) of H matrix
 - Significant singular values ($\sigma_i > 0$): use denoising process
 - Zero/small singular values: use diffusion model prediction
- SVD separates signal from noise, focuses diffusion only where no information exists
- Significantly speeds up diffusion process vs pure DDPM

Current Limitations & Research

- Assumes uncorrelated noise (identity covariance)
 - Real-world noise typically correlated between neighboring pixels
 - Correlation-aware diffusion models paper addressing this (CVPR presentation Saturday)
- Purely data-driven approach lacks physics constraints
 - No understanding of physical structures (walls can't break/reform)
 - Causality and physical understanding integration needed
- Weird artifacts when predicting over large time gaps

VGG Architecture Introduction

- Visual Geometry Group architecture presentation beginning
- Takes multiple other architectures as components

- Will cover: previous work, weaknesses, pipeline, strengths, limitations

Chat with meeting transcript: <https://notes.granola.ai/t/4e230273-a890-44c2-9ff8-040688796a8f>

These are three different ways of recovering 3D structure from images:

Method	Main Idea	Input	Output	Strength	Weakness
Shape from Shading (SfS)	Infer shape from how light falls on a surface	1 image	Surface shape/depth	Works from a single photo	Very sensitive to lighting assumptions
Structure from Motion (SfM)	Infer 3D by tracking points across many photos taken from different viewpoints	Multiple images	Sparse 3D point cloud + camera positions	Accurate geometry	Needs many overlapping images
NeRF	Learn a neural representation of a scene and render new viewpoints	Multiple images	Continuous 3D scene that can generate novel views	Photorealistic novel views	Slow training/rendering

Shape from Shading

Imagine looking at a tennis ball.

The center appears bright and the edges darker.

SfS asks:

"Given these brightness changes, what 3D shape would create them?"

It relies heavily on assumptions about lighting and material.

Good for estimating local surface curvature.

Structure from Motion (SfM)

Imagine walking around a statue and taking 100 photos.

SfM finds matching points:

- Nose in image 1
- Same nose in image 2
- Same nose in image 3

Using geometry, it estimates:

- Where the cameras were
- Where the 3D points are

Result:

Photos

↓

Feature Matching

↓

Camera Poses

↓

Sparse 3D Point Cloud

This powers many photogrammetry systems.

NeRF

NeRF starts similarly:

Many photos

↓

Estimate camera poses

↓

Train neural network

But instead of producing a sparse point cloud, it learns:

"What color and density exist at every point in space?"

Then it can render completely new viewpoints.

Example:

- 50 photos of a room
- Ask for view #51 from a never-seen angle
- NeRF generates it

Result:

Photos



Neural Scene Representation



Novel View Synthesis

Relationship

Think of them historically:

Shape from Shading (1980s)



Structure from Motion (1990s-2010s)



NeRF (2020+)

And conceptually:

- SfS: Recover shape from lighting.
- SfM: Recover shape from camera movement.
- NeRF: Learn an entire scene representation that enables new views.

Most modern NeRF pipelines actually use SfM first to estimate camera positions before training the NeRF.

Photometry and Light Reflection Models

- Light reflection fundamentals
 - Image formation combines geometry + photometry
 - Geometry determines shape/size, photometry determines pixel intensity
 - Bidirectional Reflectance Distribution Function (BRDF) models light reflection proportion
- BRDF constraints
 - Energy conservation (outgoing $\leq$ incident light)
 - Reciprocity principle (reversible light paths)
- Three types of light interaction
 - Reflection (focus of computer vision)
 - Transmission (medical imaging, X-rays)
 - Emission (thermal imaging, infrared)

Diffuse vs Specular Reflection

- Diffuse reflection (Lambertian surfaces)
 - Follows Lambert's law: emitted light $\propto \mathbf{N} \cdot \mathbf{L}$ (surface normal $\cdot$ light direction)
 - Brightness independent of viewing direction
 - Material property: albedo (kd)
 - Mathematical simplicity makes it common in CV applications
- Specular reflection
 - Intensity depends on viewing direction
 - Modeled as $(\mathbf{V} \cdot \mathbf{R})^{n_s}$ where $\mathbf{V}$ =viewing direction, $\mathbf{R}$ =reflection direction
 - Requires empirical calibration of n_s parameter
 - More realistic but mathematically complex

Shape from Shading

- Single image 3D reconstruction technique
 - Uses photometric information to estimate surface normals
 - Requires known light direction and material properties
 - Can work with only one image (vs structure from motion requiring multiple)
- Practical limitations
 - Extremely difficult due to unknown light sources
 - Light rarely comes from single point source
 - Inter-reflections complicate the model
- Successful applications
 - Space/Mars imaging where sun position is known
 - Crater detection from single satellite images
 - Very specialized, bespoke use cases only

Student Presentations

- Camera Control for Video Generation (2025 paper)
 - Plugin module for precise camera motion control in video diffusion models
 - Uses Plucker coordinates to encode 6D camera trajectory information
 - Addresses limitation where models move objects instead of camera
 - Modular design works with existing architectures (AnimateDiff, Stable Video Diffusion)
 - Applications in filmmaking for emotional storytelling through camera movement
 - Neural Radiance Fields (NeRF) - 2020 UC Berkeley/Google
 - Novel view synthesis from handful of photos (50 → 51st angle)
 - Neural network inputs: 3D position (xyz) + viewing direction
 - Outputs: RGB color + density (opacity)
 - Key techniques: positional encoding for detail, hierarchical sampling for efficiency
 - Results: 5MB storage vs 15GB for voxel grids, 25-30 training images needed
 - Limitations: 1-2 days V100 GPU training, 30 seconds per frame, breaks down far from training views
-

Chat with meeting transcript: <https://notes.granola.ai/t/5e308b25-52bc-46b0-903f-5434181682d5>

Stereo Reconstruction Geometry Fundamentals

- Two-camera baseline system with global coordinate system fixed at left camera optical center
- Translation vector t represents right camera position relative to left camera
- 3D reconstruction process:
 - Find corresponding points in both images (p and q)
 - Shoot rays from each camera center through image points
 - Ray intersection determines 3D point location
 - Requires perfect correspondence for accurate reconstruction

Epipolar Geometry and Constraints

- Epipolar plane defined by three elements:
 - Translation vector t between cameras
 - Ray from left camera through point p
 - Ray from right camera through point q
- Epipolar lines are projections of 3D rays onto opposite image planes
- All epipolar lines converge at epipoles (projections of camera centers)
- Forms “pencil of lines” structure in differential geometry

Fundamental Matrix Properties

- 3×3 matrix F that maps points in image 1 to epipolar lines in image 2
- Key relationships:
 - Epipolar line of point p : Fp
 - Epipolar constraint: $q^T Fp = 0$ for corresponding points
 - $Fe_1 = 0$ and $F^T e_2 = 0$ (epipole properties)
- Rank 2 matrix (not full rank despite being 3×3)
- Contains 8 unique parameters (9th set arbitrarily due to scale ambiguity)

Zero-Shot Novel View Synthesis (Student Presentation)

- Paper: “Zero-1-to-3” - generates new viewpoints from single input image
- Uses pre-trained Stable Diffusion with camera control training
- Key innovations:
 - Works on non-existent objects (paintings, artistic renderings)
 - No template dependency (CAD models, meshes, voxels)
 - Learns geometric relationships through millions of image pairs
- Training pipeline combines input/output views with camera rotation/translation embeddings

- NeRF reconstruction component for 3D mesh generation
- Outperformed competitors in detail retention and ground truth accuracy
- Published 2023 as first zero-shot framework for novel view synthesis

Chat with meeting transcript: <https://notes.granola.ai/t/379f7d52-2e1a-4f0b-8841-3ab329f87041>

Structure from Motion Fundamentals

- Small baseline problem using consecutive video frames
 - Assumes temporal continuity for derivative calculations
 - Reprojection error minimization through bundle adjustment
- Key limitations and dependencies
 - Translation and depth interdependent (rotation independent)
 - Scale ambiguity: cannot determine absolute scale without reference point
 - Requires minimum 5 points for solvable system ($n+5$ unknowns, $2n$ equations)
- Focus of expansion/contraction
 - Ratio of motion components (t_z/t_x)
 - Vectors converge to point (contraction) or diverge (expansion)
 - Parallel motion when $t_z=0$ creates focus at infinity

Mathematical Derivation Process

- Setup: Static scene with moving camera
 1. World coordinate system placed on object centroid
 2. Camera motion: translation vector t , rotation vector ω
 3. Relative object velocity: $-t - \omega \times p$
- 3D to 2D projection workflow
 1. Compute instantaneous 3D velocity of points
 2. Apply perspective projection equations
 3. Derive optical flow components (u,v) from time derivatives
 4. Algebraic manipulation yields structure from motion equations
- Key insight: Optical flow = projection of 3D motion onto 2D plane

Practical Implementation Considerations

- Unknowns breakdown
 - n depth values (one per point)
 - 5 motion parameters (2 translation ratios + 3 rotation)
 - Scale ambiguity reduces 6 motion parameters to 5
- Extensions for non-rigid objects
 - Linear combination of rigid basis vectors
 - Solve structure from motion for each basis separately
- Applications: Underlies COLMAP, robotics, autonomous systems

Stable Video Diffusion Overview

- Three-stage data curation pipeline
 - Image pre-training for visual understanding
 - Video pre-training with quality filters (CLIP score, aesthetic score, OCR detection, optical flow)
 - Fine-tuning on smaller high-quality dataset (LVD-F)
- Performance improvements
 - 16-hour training vs 6-day competitor training
 - Higher FVD scores and human preference ratings
 - Multi-view capabilities through fine-tuning
- Open challenges: AI-generated video detection and continual learning for evolving generators

Chat with meeting transcript: <https://notes.granola.ai/t/47f24dbd-9a34-4694-93ae-9a680213b9e7>

3D Reconstruction Overview

- Fundamental equation: $x = \pi \times X$
 - π = calibration matrix (intrinsics + extrinsics)
 - Extrinsics = rotation R and translation (T) of camera relative to world coordinate system
 - x = 2D image points, X = 3D world points
- Course progression: representation $\rightarrow$ camera modeling $\rightarrow$ 3D reconstruction
- Two main reconstruction methods: Structure from Motion (SfM) and Stereo

Structure from Motion Fundamentals

- Input: video sequence or image series with relative motion between camera and scene
- Works with single moving camera vs static scene OR static camera vs moving scene
- Small baseline system (consecutive frames have minimal motion)
 - Uses continuity assumptions and brightness constancy
 - Contrasts with stereo (wide baseline, large motion between views)
- Foundation for COLMAP and other 3D reconstruction tools
- Applications: point cloud generation, camera pose estimation

Mathematical Framework & Key Properties

- Input: 2D points with correspondences ($p_i \leftrightarrow$ ith point in jth frame)
- Output: 3D locations (X_i) and camera parameters ($R \square, T \square$)
- Optimization via bundle adjustment
 - Minimizes reprojection error between observed and projected points
 - Nonlinear optimization (Levenberg-Marquardt, gradient descent, neural networks)
- Critical limitations from scale ambiguity:
 - Translation affects depth; pure rotation does not
 - Cannot estimate translation and depth simultaneously without reference
 - Solution: estimate ratios ($Z/T_z =$ inverse scaled depth, $T_x/T_z =$ focus of expansion/contraction)

Course Schedule & Presentation

- Remaining structure: Friday (SfM math derivation), Monday holiday, Wed/Fri (stereo)
 - Final topics: illumination/lighting, generative models (VAE $\rightarrow$ diffusion models)
 - Student presentation on VL3 multimodal model evolution
 - Progression: VL2.5 $\rightarrow$ VL3 $\rightarrow$ VL Candidate
 - Key advancement: longer context windows (257k tokens)
 - Mixture of experts architecture for agentic systems
 - Reasoning/planning capabilities for module selection
-

Chat with meeting transcript: <https://notes.granola.ai/t/c94efeed-5af6-4754-a493-08df3ce02a1e>

Camera Calibration Methods

- Direct mathematical approach (black box)
 - Solves homogeneous system $Ax = 0$ using eigenvector of minimum eigenvalue
 - Scale ambiguity problem: any scalar multiple is also valid solution
 - Works mathematically but doesn't separate intrinsic/extrinsic parameters
 - Cannot determine focal length, camera pose individually
- Step-by-step physical approach
 - Separates rotation matrix (orthonormal), translation, projection, intrinsics
 - Uses orthonormality constraints to solve for individual camera parameters
 - More complex algebra but provides interpretable results
 - Enables understanding of specific camera properties

Scale Ambiguity in 3D Vision

- Fundamental physics limitation across all 3D reconstruction
 - Any point on projection ray creates same image
 - Translation + proportional scaling = identical projection
 - Scale ambiguity arises from translation, not rotation
 - Will appear in structure from motion, stereo vision
- Requires reference point solution
 - Need known 3D position for at least one scene point
 - Reference establishes absolute scale for entire reconstruction
 - Without reference, object could be anywhere along projection ray

Homography for Planar Surfaces

- Linear transformation between corresponding points on planar surfaces
 - $x' = Hx$ for points lying on same plane
 - Useful for security cameras, rooftop surveillance
 - Valid approximation when objects far from camera (height $\ll$ distance)
- Solution requirements
 - Need minimum 4 point correspondences
 - 9 unknowns but scale ambiguity reduces to 8 dimensions
 - Solve using eigenvector of $A^T A$ for minimum eigenvalue
 - More points enable least squares fitting for better accuracy

Vision-Language Models (LLaVA)

- Architecture combines vision encoder + language model
 - Projects image features to language embedding space
 - Enables visual question answering, image captioning
 - Can handle rotated images, demonstrates robustness
 - Grounding improves performance significantly
 - Direct VLM queries often inaccurate
 - Better results when feeding detector outputs (people, vehicles, signs) to VLM first
 - Similar to RAG architecture but for input processing rather than output generation
-

Chat with meeting transcript: <https://notes.granola.ai/t/59815620-ec84-49fe-84ed-94aaf410e540>

Projective Geometry & Camera Calibration Overview

- Parallel lines on ground plane converge to vanishing point at infinity
- Different parallel line sets create different vanishing points forming horizon
- Goal: 3D reconstruction from 2D images requires understanding complete image formation process

Camera Coordinate Systems & Parameters

- Two coordinate systems needed:
 1. Camera coordinate system (moving with camera)
 2. World coordinate system (fixed reference frame)
- Parameter types:
 1. Intrinsic parameters
 - Focal length
 - Pixel scaling ratios
 - Optical axis offset (often set to zero)
 2. Extrinsic parameters
 - Camera rotation and translation relative to world system
- Forward model: 3D world → camera coordinates → projection → image plane
- Inverse problem: Given 2D pixels, estimate 3D structure (requires calibration first)

Direct Calibration Method

- Treats entire transformation as black box matrix (3×4)
- Mathematical approach without physical parameter separation
- Process:
 1. Set up equations: small $x = \pi \times \text{capital } X$
 2. Cross multiply to eliminate homogeneous coordinates
 3. Reorganize into linear system: $Ax = 0$
- Homogeneous system challenges:
 1. Infinite solutions due to scale ambiguity
 2. Cannot solve $Ax = 0$ uniquely
- Solution approach:
 1. Minimize $\|Ax\|^2$ subject to $\|x\|^2 = 1$
 2. Use Lagrange multipliers for constrained optimization
 3. Results in eigenvalue problem: $A^T A x = \lambda x$
 4. Take eigenvector corresponding to smallest eigenvalue

Calibration Implementation & Limitations

- Calibration setup uses checkerboard patterns for precise 2D point detection
 - Scale ambiguity fundamental to projection geometry
 1. Any point on ray projects to same image location
 2. Requires external reference to resolve scale
 - Direct method limitations:
 1. No physical insight into camera pose or intrinsics
 2. Cannot incorporate known parameters (e.g., focal length from spec sheet)
 3. Doesn't model radial distortion
 4. Error function may not be optimal for physical problem
 - Alternative: Separate estimation of intrinsic/extrinsic parameters (more complex but physically meaningful)
-

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Camera Models and Perspective Projection

- Perspective projection most important/realistic camera model
- Image height = focal length $\times$ (3D Y-coordinate / Z-coordinate)
- All measurements relative to camera's center of projection
- Camera coordinate system: X into page, Y up, Z along optical axis
- Optical axis passes through center of projection
- Image plane positioned at focal distance

Intrinsic vs Extrinsic Camera Parameters

- Intrinsic parameters (camera-specific, unchanging):
 - Focal length (f)
 - Pixel scaling factors (s_x, s_y) - converts millimeters to pixels
 - Principal point - where optical axis intersects image plane
- Extrinsic parameters (camera pose in world):
 - Rotation matrix $\mathbf{R}$ and translation vector ($\mathbf{t}$)
 - Transforms world coordinates to camera coordinates
 - Critical for moving cameras/robotics applications

Mathematical Framework and Calibration

- Core equation: $\mathbf{x} = \mathbf{P} \times \mathbf{X}$ (2D image point = projection matrix $\times$ 3D world point)
- Encapsulates "all of computer vision"
- Uses homogeneous coordinates for linear transformations
- Calibration scenarios:
 1. Calibrated: Known camera parameters, estimate pose/3D structure
 2. Uncalibrated: Estimate both camera parameters and scene structure
- Transformation pipeline: World $\rightarrow$ Camera $\rightarrow$ Projection $\rightarrow$ Image pixels

Projective Geometry and Scene Graphs

- Perspective projection properties:
 - Straight lines remain straight
 - Parallel lines converge at vanishing points
 - All vanishing points on a plane form the horizon line
- Scene graph generation (SGG):
 - Represents images as graphs: nodes = entities, edges = relationships
 - New transformer-based approach treats as bipartite graph problem
 - Outperforms traditional two-stage detection methods
 - Applications: image captioning, synthesis, video analysis

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Camera Projection Models

- Perspective projection: fundamental 3D to 2D transformation
 - Uses similar triangles: $x = f(X/Z)$, $y = f(Y/Z)$
 - f = focal length (intrinsic camera parameter)
 - Creates nonlinear relationship making 3D reconstruction challenging
- Orthographic projection: $x = X$, $y = Y$
 - Parallel light rays, no depth scaling
 - Works when objects very far from camera
 - Cannot solve for depth information
- Weak perspective (scaled orthographic): introduces constant scale factor
 - Valid when object depth variation $\ll$ distance from camera
 - Example: person imaging from surveillance camera
 - Breaks down when object far from optical axis

Homogeneous Coordinates & Linear Systems

- Transform nonlinear projection into linear matrix operations
 - 2D point (x,y) becomes $(x,y,1)$ in homogeneous coordinates
 - 3D point (X,Y,Z) becomes $(X,Y,Z,1)$
- Perspective projection matrix enables linear algebra solutions
 - Allows least squares, matrix inversion techniques
 - Similar to kernel SVM lifting to higher dimensions
- Scale ambiguity problem: infinite points on ray project to same image point
 - Third coordinate serves as scale parameter
 - Fundamental challenge in 3D reconstruction

Coordinate System Transformations

- Three reference systems required for complete modeling:
 - World coordinate system (fixed reference)
 - Camera coordinate system (origin at camera center)
 - Image coordinate system (pixel measurements)

- Intrinsic parameters: transform camera to image coordinates
 - Focal length, pixel size (sx, sy)
 - Internal to camera hardware
- Extrinsic parameters: transform world to camera coordinates
 - Camera pose estimation critical for 3D reconstruction
 - Moving cameras require world reference system

Research Presentation: ConvLoRA for SAM

- Domain adaptation problem: SAM struggles with medical/satellite imagery
 - Lacks spatial locality understanding
 - Trained only for binary segmentation masks
 - ConvLoRA solution: adds lightweight convolution experts to LoRA
 - Mixture of experts for multi-scale feature extraction
 - Captures local patterns, edges at different scales
 - Fine-tunes only 0.63% of total SAM parameters (4.02M)
 - Results across 4 domains show significant improvement
 - Medical, agricultural, remote sensing applications
 - Maintains computational efficiency while adding spatial awareness
-

Chat with meeting transcript: <https://notes.granola.ai/t/2ead89a2-1971-44df-8f67-c6617fb21267>

Electromagnetic Spectrum & Image Formation

- Visible light only small part of electromagnetic spectrum
- Beyond visible: infrared regions, radio waves, microwaves
 - Each wavelength interacts differently with surfaces
 - Night imaging requires infrared (thermal has own properties)
- Course focus: primarily visible spectrum with some beyond-visible coverage

Camera Physics & Optics

- Basic pinhole camera creates blurred images due to light scattering
- Aperture controls light amount but reduces brightness
 - Too small aperture → diffraction effects (quantum mechanics limit)
 - Optimal aperture balances sharpness vs brightness
- Lens addition enables light focusing while capturing more rays
 - Allows sharp focus at specific distances
 - Introduces depth of field concept
- Thin lens equation: $1/v_o + 1/d_i = 1/f$
 - v_o = object distance, d_i = image distance, f = focal length

3D Reconstruction Challenges

- Computer vision = inverse problem (image → 3D scene reconstruction)
- Key parameters to estimate:
 - Camera intrinsic parameters
 - 3D object geometry
 - Camera pose (position & orientation)
- Reference frame critical for distance measurements
 - Camera-centric vs world-centric coordinates
 - Static camera → world reference preferred
 - Dynamic camera → camera reference essential
- Pinhole camera model: origin at center of projection
 - X,Y axes aligned with image plane
 - Z axis perpendicular to image plane

Vision Transformer Self-Supervision (Student Presentation)

- Banner 3 model: self-supervised vision transformer
- Motivation: replicate NLP pre-training success in vision
- Dataset: 1.2B web images → curated to 142M after deduplication
- Teacher-student architecture:
 - Teacher: stable network with multiple crop views
 - Student: matches teacher output through learning
- Results: 2x faster, 3x less memory than previous models

- Real-world application: North American forest classification
- Limitation: bias toward Western imaging data

Chat with meeting transcript: <https://notes.granola.ai/t/d8e8b75e-76f2-4e22-be9f-b33384a2e2b5>

Machine Learning Approaches in Computer Vision

- Traditional vs modern approaches
 1. Traditional: hand-crafted features + separate classification
 2. Modern: end-to-end training of feature extraction + classification
- Learning paradigms covered:
 1. Supervised learning - examples with ground truth, minimize loss function
 2. Unsupervised learning - clustering costs, similar features cluster together
 3. Self-supervised learning - uses data structure as supervisory signal
 - Example: jigsaw puzzles where colors/shapes must match
 4. Reinforcement learning - state/action/reward framework
 - Policy learning: action given state
 - Trial and error process, 60-70 years old but now popular with deep learning
 5. Imitation learning - no reward function, match expert demonstrations
 - Often combined with RL for faster training

Domain Adaptation and Continual Learning

- Core challenge: performance degrades when applying models to different domains
- Common domain gaps:
 - Weather conditions, indoor/outdoor, day/night
 - RGB to thermal cameras
 - Simulation to real world deployment
- Catastrophic forgetting problem
 - Models lose previous domain performance when adapting to new domains
 - Major research challenge in continual learning
- Distribution mismatch between training and inference data

Context in Computer Vision Recognition

- Spatial context importance
 - Objects appear in expected environments (laptop with table/chair, not forest)
 - Example: boat vs airplane disambiguation using grass context
- Temporal context for activity recognition
 - Sequential actions provide recognition cues
 - Person opening car door → trunk → activity sequence
- Scene graphs for contextual representation
 - Objects: man, horse, bucket, glasses
 - Relationships: man wearing glasses, feeding horse, holding bucket
 - Applications: image captioning, image generation, grounding for LLMs
- Dynamic scene graphs evolve over time
 - Richer description than simple captions
 - Better grounding prevents model hallucination

Performance Measurement Metrics

- Confusion matrix fundamentals:
 - True positives, true negatives, false positives, false negatives
 - Type 1 errors (false positives), Type 2 errors (false negatives)
- Key metrics definitions:
 - Accuracy = $(TP + TN) / (TP + TN + FP + FN)$
 - Recall (True Positive Rate) = $TP / (TP + FN)$
 - Precision = $TP / (TP + FP)$
 - F1 Score = harmonic mean of precision and recall
- ROC curves and operating points
 - Plot true positive rate vs false positive rate
 - Vary thresholds to generate curve points
 - Equal error rate: FPR = FNR for balanced operation
 - Area under curve indicates system quality
- Cumulative Match Characteristic (CMC)
 - Measures how quickly correct results appear in ranked lists

- Important for search/retrieval systems
- Accounts for result ordering, not just presence

CLIP Model Presentation

- Contrastive Language-Image Pre-training by OpenAI (2021)
- Training approach:
 - 400M image-text pairs from internet
 - Image encoder: ResNet or Vision Transformer
 - Text encoder: Transformer model
 - Contrastive loss: similar pairs close, different pairs apart
- Zero-shot classification capabilities
 - No training on specific datasets required
 - Prompt engineering affects performance significantly
 - "Photo of a blue sports car" > "photo of a car" for specificity
- Performance characteristics:
 - Comparable to ResNet-50 in zero-shot setting
 - More robust to natural distribution shifts
 - Struggles with fine-grained tasks (MNIST variants)
 - Cannot handle spatial relationships (dog on bike vs bike on dog)

Chat with meeting transcript: <https://notes.granola.ai/t/0f418c24-710b-41d3-a782-b6d64dda8367>

Machine Learning Fundamentals for Computer Vision

- Two primary ML problem types:
 - **Regression:** Continuous world states (pose estimation, head pose)
 - **Classification:** Discrete world states (face detection, pedestrian detection, segmentation)
- Core ML setup: Compute probability of world state W given data X
 - X = observed data (images/videos)
 - W = world state inferences (camera motion, face identity, 3D structure)
 - Challenges: noisy data, inherent visual ambiguity

Discriminative vs Generative Models

- **Discriminative models:** Compute $P(W|X)$
 - Direct inference from data to world state
 - Examples: classifiers, pose estimators
 - Easier inference - direct evaluation
- **Generative models:** Compute $P(X|W)$
 - Generate data given world representation
 - Examples: diffusion models, interpolation systems
 - Require Bayes theorem for inference: $P(W|X) = P(X|W) \times P(W) / P(X)$
 - Need prior knowledge $P(W)$ for world-grounded generation

Computer Vision Problem Examples

- **Regression applications:**
 - Pose estimation from silhouettes
 - Head pose estimation
 - Fundamentally continuous variables
- **Classification applications:**
 - Face detection (binary: present/absent)
 - Face recognition (multi-class discrete)
 - Semantic segmentation (pixel-level class labels)
 - Background subtraction (foreground/background binary)

Learning Paradigms Overview

- **Supervised learning:** Labeled training pairs (X_i, W_i)
- **Unsupervised learning:** No labels
 - Clustering, anomaly detection
 - PCA as dimensionality reduction example
- **Self-supervised learning:** Structure-based supervision
 - Jigsaw puzzles, token prediction (LLM training approach)
 - Natural data relationships as supervisory signal

- **Reinforcement learning:** Reward-based policy learning
 - State-action-reward framework
 - Used in robotics and RLHF for foundation models
 - **Imitation learning:** Learn from expert demonstrations
 - Policy matching from video observations
 - Warehouse/household robot training applications
-

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Kalman Filter Fundamentals

- Recursive algorithm for tracking objects over time
 - Combines motion model predictions with detector observations
 - Uses weighted combination based on uncertainty (standard deviations)
- Two-step process:
 - Prediction: propagate previous estimate using motion model
 - Correction: adjust prediction using new observation
- Kalman gain determines weighting
 - High prediction noise → more weight to observation
 - High measurement noise → more weight to prediction
- Innovation process captures difference between observation and prediction
 - New information added to prediction
 - Zero difference means perfect motion model

Bayesian Tracking Approach

- Combines prior knowledge with observations
 - Prior: motion model understanding
 - Observations: detector outputs
- Foundation for reasoning frameworks in AI/ML
 - Agentic systems use similar prior + observation combination
 - General approach beyond just tracking applications
- Multi-dimensional extension uses covariance matrices
 - Same principles as 1D case
 - Handles position vectors, multiple features

Data Association Problem

- Critical challenge: matching predictions to detections across frames
 - Multiple objects create ambiguity
 - Need to determine which detection corresponds to which prediction
- Solutions involve feature matching
 - Color, position, appearance features
 - Deep network-based matching algorithms
- Related to re-identification problem
 - Learning transformations between camera views
 - Siamese networks or reconstruction approaches
 - Training on known correspondences, applying to test data

Particle Filter Alternative

- Represents uncertainty as weighted particle distribution
 - Particles spread across image with varying weights
 - High weights near object, low weights elsewhere
- Advantages for occlusion handling
 - Maintains particles throughout image during occlusion
 - Can recover object when it reappears
 - Continuous search over entire image space
- Major disadvantage: computational cost
 - Requires particles everywhere to enable recovery
 - Much higher computation than bounding box tracking

Paper Presentations Summary

- RAFT (Optical Flow)
 - All-pairs correlation volume for motion estimation
 - Recurrent GRU updates at fixed resolution
 - 30% error reduction on Sintel dataset
 - Trained on synthetic data (ground truth motion vectors impossible for real data)
 - ByteTrack (Multi-Object Tracking)
 - Associates low-confidence detections with existing tracklets
 - Prevents loss of partially occluded objects
 - 30 FPS performance with significant accuracy improvements
 - Limitation: struggles with long-term occlusions (30+ frames)
-

Chat with meeting transcript: <https://notes.granola.ai/t/13facdac-43f5-43de-8ea2-07a7aafdd24a>

Bayesian Tracking Fundamentals

- Mathematical foundation for object tracking using prediction-correction approach
- Two key simplifying assumptions enable tracking:
 - Markovian property - current state depends only on previous state, not entire history
 - Conditional independence - measurements depend only on current state
- Bayes' theorem links conditional probabilities: $P(A|B) = P(B|A) \times P(A) / P(B)$
 - Foundation for all data science and communications systems
 - Enables estimation of transmitted signals from received observations

Prediction-Correction Process

- Prediction step: estimate object location at time i using observations up to $i-1$
 - Uses motion model (constant velocity, constant acceleration)
 - Incorporates system dynamics: x_i given x_{i-1}
 - No new observations used in this step
- Correction step: refine prediction using actual observation y_i
 - Combines prior knowledge with new measurement
 - Weighted combination based on uncertainty levels
 - Higher uncertainty $\rightarrow$ lower weight in final estimate

Kalman Filter Implementation

- Special case of Bayesian tracking with linear models and Gaussian noise
- Key equations for 1D case:
 - Prediction: $\hat{x}_i^- = A \times \hat{x}_{i-1}^+$ (propagate previous estimate)
 - Correction: weighted combination of prediction and measurement
- Kalman gain determines weighting:
 - Low prediction uncertainty $\rightarrow$ trust prediction more
 - Low measurement uncertainty $\rightarrow$ trust observation more
- Innovation process: difference between prediction and actual observation
 - Represents new information available for correction

Practical Applications & Considerations

- Drift problem: prediction-only tracking accumulates errors over time
 - Similar to model parameter drift in continuous learning systems
 - Correction step prevents error accumulation
 - Motion models examples:
 - Constant velocity: $u_i = u_{i-1} + \text{noise}$
 - Constant acceleration: includes acceleration term
 - Non-Bayesian vs Bayesian approaches:
 - Non-Bayesian: detection + association only
 - Bayesian: combines prior knowledge with observations for better performance
-

Chat with meeting transcript: <https://notes.granola.ai/t/cd207378-050d-4055-9111-76cfccd32bb4>

Optical Flow Fundamentals

- Brightness constancy assumption
 - Intensity remains constant in image patches
 - Some change occurs in practice
- Cannot compute optical flow for single pixel
 - Requires patch-based computation for regularization
 - Provides multiple equations to solve flow vectors
- Aperture problem
 - Fundamental computer vision limitation
 - Small aperture viewing prevents determining actual motion direction
 - Can only detect normal flow (projection along gradient)
 - Need corner features or texture for true flow computation

Image Warping vs Filtering

- Filtering changes pixel values (range of function)
- Warping changes spatial coordinates (domain of function)
 - Stretching, rotation, aspect ratio changes
 - Essential for modeling camera-scene relative motion
- Applications in device compatibility (computer to phone display)

Motion Transformation Types

- Translation: 2 degrees of freedom (tx, ty)
- Euclidean: Translation + rotation (3 DOF)
- Similarity: Translation + rotation + scaling (4 DOF)
- Affine: Translation + rotation + scaling + shear (6 DOF)
- Perspective: 3D to 2D projection (8 DOF)
 - Required for out-of-plane rotation
 - Uses homogeneous coordinates
 - Handles scale ambiguity in 3D reconstruction

Course Direction & 3D Reconstruction

- Second half focuses on 3D reconstruction
 - Current frontier in AI/physical world integration
 - Critical for autonomous systems and robotics
- Perspective projection geometry
 - Ray from optical center through 3D point to image plane
 - Scale ambiguity: infinite points on same ray project to same image point
 - Homogeneous coordinates represent this mathematically
- Parametric motion models enable constrained optical flow
 - 6 parameters for affine motion vs 2 per pixel
 - Solvable system: 10,000 equations, 6 unknowns for 100x100 image

Chat with meeting transcript: <https://notes.granola.ai/t/8ad5ad9d-ecb7-40ba-8d1e-d5fc52e6a23e>

Motion Estimation Fundamentals

- Transition from single images to image sequences/videos
- Three methods to cover:
 - Patch-based motion estimation
 - Image warping
 - Parametric/global motion estimation
- Motion critical for:
 - Activity recognition
 - 3D reconstruction (nearly impossible from single image without strong priors)

Optical Flow Theory

- Brightness constancy assumption: patch brightness remains constant across frames
 - Fundamental requirement for optical flow computation
 - Only works with small motion between consecutive frames
- Mathematical foundation:
 - $I(x,y,t) = \text{constant over time}$
 - Leads to: $I_x \cdot u + I_y \cdot v + I_t = 0$
 - Two unknowns (u,v motion vector), one equation per pixel
- Aperture problem:
 - Cannot solve unique motion from single equation

- Only computes “normal flow” - projection of actual motion onto image gradient
- Geometric limitation when viewing edges through small apertures

Patch-Based Solution Approach

- Regularization through spatial consistency:
 - Assume neighboring pixels share same motion vector (u,v)
 - Creates more equations than unknowns within window
 - Enables least squares solution
- Matrix invertibility requirements:
 - Full rank gradient matrix needed
 - Fails in textureless regions (no gradients)
 - Fails along single edges (gradients in one direction only)
- Window size trade-offs:
 - Larger windows blur object boundaries
 - Smaller windows insufficient for robust estimation

Optical Flow Visualization & Logistics

- HSV representation standard:
 - Hue indicates motion direction
 - Saturation shows motion magnitude
 - Examples: red for rightward motion, blue for leftward
- Next week schedule uncertainty due to jury duty
 - Check Canvas messages before class
 - Will post daily updates on class status

Chat with meeting transcript: <https://notes.granola.ai/t/a5f37e90-52f2-4f25-9437-3d6b38bfc88>

Feature Detection Fundamentals

- Registration problem: finding matching points between images
 - Exceptionally difficult due to lighting shifts, similar features across scenes
 - Takeo Kanade's famous statement: “There are three problems in vision: registration, registration, registration”
 - 3D reconstruction mathematically solvable if features match properly
- Good features require two types of invariance:
 - Geometric invariance - robust to camera/object movement
 - Photometric invariance - robust to lighting changes
- Local features preferred over global features
 - Robust to occlusion and clutter
 - Many distinctive points available
 - Global features fail when portions occluded

Mathematical Framework for Feature Quality

- Cost function: sum of squared errors between image patches
- Taylor series approximation for small motions leads to matrix H:
 - $H = [\Sigma(I_x^2), \Sigma(I_x I_y); \Sigma(I_x I_y), \Sigma(I_y^2)]$
 - Resembles covariance matrix of image gradients
- Principal component analysis on gradient directions:
 - Flat regions: no principal directions, poor features
 - Edges: one significant direction
 - Corners: two equal principal directions, best features
- Harris corner detector: $\lambda_1 \lambda_2 / (\lambda_1 + \lambda_2)$
 - Uses determinant/trace ratio to avoid expensive eigen analysis
 - Still used in robotics for low-compute applications

SIFT Features and Matching Challenges

- SIFT (Scale-Invariant Feature Transform):
 - Computes histogram of gradients in 4x4 regions
 - Matches features with similar histograms in same image region
 - Good robustness to geometric/photometric variations
- Extreme matching difficulties:
 - Tiny object detection (prairie dogs, satellite cars) achieves only 50-60% accuracy
 - Camouflaged objects nearly impossible even for state-of-art methods
 - Objects <30 pixels wide lack distinguishable features
- Alternative imaging modalities (thermal) limited by:

- Expensive, bulky equipment
- Scale deployment challenges

Evaluation and System Design

- ROC curves measure true positive vs false positive rates
 1. Originally developed for WWII radar aircraft detection
 2. Area under curve compares different methods
 3. Operating point chosen by varying distance thresholds
 - Critical data separation:
 1. Training data - build model
 2. Validation data - compute ROC curves, choose operating point
 3. Test data - final evaluation (never seen during development)
 - Cross-validation: randomly sample validation sets 5 times for error bars
 - Distance vs similarity metrics interchangeable with threshold adjustment
-

Chat with meeting transcript: <https://notes.granola.ai/t/1391440f-95f6-4bbc-816a-b66b376e31a5>

Deformable Contours (Active Contours/Snakes)

- Segmentation technique using iterative contour deformation
 - Start with initial contour (e.g., circle)
 - Deform to match object boundaries
 - Stop when energy function minimized
- Energy function combines two components:
 - External energy - matches contour to image edges
 - Uses negative gradient magnitude (minimization process)
 - Attracts contour to edge valleys in 3D intensity surface
 - Internal energy - maintains shape smoothness
 - First derivative measures elasticity
 - Second derivative measures curvature
- Applications and limitations:
 - Excellent for medical imaging (liver, organs, tumors)
 - Poor for complex natural images (dogs, faces)
 - Struggles with non-convex shapes due to smoothness constraints

Texture Representation Using Gabor Filters

- Texture cannot be represented using edge detection
 - Requires basis function approach similar to PCA/Fourier
 - Uses orthonormal filter banks for decomposition
- Gabor filters structure:
 - Combination of sine/cosine waves modulated by exponential decay
 - Different orientations and frequencies capture texture characteristics
 - 8 common filter orientations shown in examples
- Mathematical foundation identical to vector projections:
 - Project image patches onto basis filters
 - Weighted combination reconstructs original texture
 - Same principle as 2D vector decomposition but higher dimensions

Deep Network Explainability

- Class Activation Maps (CAM) and GradCAM techniques
 - Backpropagate gradients selectively to identify activating regions
 - Shows which image areas triggered specific classifications
- Validation approach:
 - Cat classification should activate on cat regions
 - Dog classification should activate on dog regions
 - Confirms network learning correct features vs. spurious correlations
- Applications for downstream tasks:
 - Object detector validation
 - Feature relevance verification
 - Part-based analysis (wings for birds, rackets for tennis)

Network Architecture Considerations

- Parameter vs. data relationship critical:
 1. Vision transformers need billions of parameters

2. Require massive datasets to avoid overfitting
 3. Small data + large networks = infinite solutions problem
- Design tradeoffs:
 1. Feature design → architecture design shift
 2. Loss function optimization remains art vs. science
 3. Hand-designed filters vs. learned features
 - Hand-designed: limited coverage, not task-optimized
 - Learned: requires labeled data, prone to overfitting without sufficient samples

Course Progression

- Completed static image representations:
 - Filtering and frequency response
 - Edge detection and PCA
 - Segmentation and texture analysis
- Next week: Multiple image processing
 - Two-image analysis techniques
 - Image sequence processing methods

Chat with meeting transcript: <https://notes.granola.ai/t/f1ea33be-41ea-4d8d-a95f-6719f7256c36>

Segmentation Types Overview

- Semantic segmentation: objects grouped by class labels
 - All cars same color, segmented together
- Instance segmentation: every individual object separated
 - Each car gets own segment regardless of class
- Panoptic segmentation: combines semantic + instance approaches
- Applications vary based on segmentation type chosen

Color-Based Segmentation Approach

- Segmentation becomes clustering problem when using color features
- Process involves plotting histogram of image colors
 - Identify peaks representing different color clusters
 - Set thresholds between clusters for segmentation
- Example: two-cluster approach (black/white regions)
 - Could extend to multiple clusters (white, gray, intermediate tones)

Key Clustering Questions

- Number of clusters (K): how many regions to segment?
 - Why choose 2 vs 3+ clusters for given image?
- Cluster boundaries: where to set thresholds?
 - Should clusters be narrow or broad?
 - How to handle noise vs actual image content?
- Cluster size variation: clusters don't have uniform width
 - Some may be very narrow, others broad

K-Means Clustering Introduction

- Standard algorithm for finding cluster boundaries
- Goal: partition data points into distinct clusters (C1, C2, C3, etc.)
- Visual example shows clear blue, orange cluster separation
- Challenge: determining optimal cluster boundaries automatically
- Leads into expectation maximization algorithm discussion

Chat with meeting transcript: <https://notes.granola.ai/t/82e04ae5-7d89-4982-ae9a-f874b73e1317>

Transformer Architecture Review

- Query-key-value mechanism drives transformer training
 - Query finds matching keys from training data via projection ($Q \cdot K^T$)
 - Values associated with best matches create attention weights
- Training uses masking structure

- Mask words in sentences, predict next token
 - Network learns by maximizing attention to correct predictions
- Attention patterns evolve during training
 - Initially: self-attention only (diagonal patterns)
 - Deeper layers: cross-word correlations emerge
 - Example: "it" correctly links to "animal" in context

Vision-Language Models & Zero-Shot Learning

- Contrastive training approach
 - Text encoder + image encoder create latent representations
 - Positive pairs (puppy + "puppy") brought together
 - Negative pairs (tiger + "puppy") pushed apart
- Zero-shot capabilities limitations
 - Works because models trained on massive internet-scale data
 - May violate train/test separation principle
 - Success often due to memorized similar features, not true zero-shot

Large Model Challenges

- Catastrophic forgetting
 - Adapting to new data causes performance drop on previous datasets
 - Performance falls dramatically, not gradually
- Hallucinations in generative models
 - Models trained to always generate output
 - Example: Robot instructed to wash clothes generates washing machine actions even when none present
- Security vulnerabilities
 - Jailbreaking: forcing models to provide harmful outputs
 - Deepfakes: generating/manipulating realistic images
 - Detection and prevention remain open problems

Segmentation Fundamentals

- Gestalt principles define good segments
 - Proximity, similarity, continuation, closure, common fate
 - Concepts remain subjective and task-dependent
- Graph-based approach (Geman & Geman 1984)
 - Each pixel = node, edges weighted by similarity
 - Cost inversely proportional to similarity
 - Segmentation via graph cuts (minimize edge costs)
 - Normalized cuts prevent isolated point issues

Segmentation Types & Evaluation

- Three main approaches:
 - Semantic segmentation - all objects of same class grouped
 - Instance segmentation - each object instance separated
 - Panoptic segmentation - combines semantic + instance
- Evaluation metrics
 - Intersection over Union (IoU) for semantic
 - F1 score (harmonic mean of precision/recall) for panoptic

ResNet Paper Presentation

- Degradation problem: deeper networks show higher training error
- Residual learning framework
 - Learn residual $F(x)$ instead of direct mapping $H(x)$
 - Output = $F(x) + x$ via skip connections
 - If optimal mapping is identity, model pushes $F(x)$ to zero
- Results: Top-5 error reduced from 15.2% (AlexNet) to much lower with ResNet

Chat with meeting transcript: <https://notes.granola.ai/t/a332dc15-21aa-4130-8361-0dcc931be80c>

Neural Network Fundamentals & Philosophy

- Shift from handcrafted features to end-to-end learning
 - Traditional approach: manually design features for specific tasks
 - Modern approach: neural networks learn optimal features for given tasks

- End-to-end learning can be taken to extremes (e.g., Tesla's autonomous driving)
 - Creates "black box" problem with sparse feedback
 - Only know if output is right/wrong, not internal reasoning
- Feature hierarchy in deep networks
 - Initial layers: capture low-level features (edges, basic patterns)
 - Middle layers: detect shapes and structures
 - Deeper layers: recognize complete object configurations
 - Compositional approach builds complexity progressively

Network Architecture Components

- Convolution operations
 - Fundamental to computer vision since course inception
 - Applies filters to weight different parts of data
 - Similar to Gaussian filtering but with learned weights
 - Advantage: acts as intelligent filter based on task requirements
- Pooling mechanisms
 - Combines outputs into lower dimensions
 - Accumulates fundamental features into complex ones
 - Enables progression from edges → shapes → complete objects
- Dropout technique
 - Randomly drops weights during training
 - Prevents overfitting by not treating all features equally
 - Improves model generalization

Learned vs Handcrafted Features

- AlexNet filter analysis reveals familiar patterns
 - Learned filters show high/low frequency characteristics
 - Orientation-specific responses (similar to hand-designed features)
 - Band-pass filter properties emerge naturally
 - Network learns features similar to traditional computer vision approaches
- Interpretability challenges
 - Understanding why networks make specific decisions remains difficult
 - Unlike physics equations, neural network reasoning is opaque
 - Active research area with PhD students working on AI-generated video detection

Network Types & Training Approaches

- Supervised vs Unsupervised Learning
 - Supervised: dog/cat classification with labeled data
 - Unsupervised: PCA (already covered), clustering, dimensionality reduction
 - PCA example: eigenanalysis of covariance matrix produces feature basis vectors
- Bias-variance tradeoff
 - Larger networks require more data to avoid overfitting
 - Example: 2 points perfectly fit line, but poor generalization
 - Internet-scale data needed for complex modern architectures
- Recurrent Neural Networks
 - Allow feedback within hidden layers
 - State depends on current input + previous state
 - Enable modification of earlier learned features based on later insights
 - Difficult to train in practice

Advanced Architectures

- Autoencoders for unsupervised learning
 - Input → latent space → reconstruction
 - Loss function compares original vs reconstructed data
 - Useful when labeled data scarce (medical images, autonomous driving)
 - Can extend to action prediction rather than just reconstruction
- Transformer networks
 - Core concept: Query-Key-Value attention mechanism
 - Query projection onto stored keys ($Q^T \times K$)
 - Softmax determines relevance weights
 - Values provide final attended information
 - Training through masking strategy (predict next word)
 - "Attention is all you need" paper title

Vision-Language Models

- Multimodal representations

- Image encoder + text encoder learn joint features
- Training on co-occurring image-text pairs
- CLIP model enables text queries to retrieve relevant images
- Decoder can reconstruct images from learned representations
- Foundation for modern diffusion models

Chat with meeting transcript: <https://notes.granola.ai/t/eb694fe2-8df1-4ea4-b59d-888e57ca3557>

Principal Component Analysis (PCA) Theory

- Dimensionality reduction for feature extraction from images
 - Raw pixel vectors contain too much uninformative data
 - Need to identify informative vs non-informative features
 - Example: edge pixels more informative than uniform background pixels
- Covariance matrix structure captures data variation
 - Multi-dimensional equivalent of variance in 1D data
 - Formula: $\text{Cov}(x) = E[(x - \mu)(x - \mu)^T]$
 - Symmetric $n \times n$ matrix for n -dimensional data
- Eigenanalysis reveals principal directions
 - Eigenvectors show directions of maximum variance
 - Eigenvalues indicate importance of each direction
 - Higher eigenvalues = more significant variation

PCA Implementation Process

- Sample statistics vs population statistics
 1. Cannot compute true expectation - use sample approximation
 2. Sample mean: $(1/n)\sum x_i$ approximates true mean when $n \rightarrow \infty$
 3. Sample covariance matrix computed from available data points
- Mathematical transformation steps:
 1. Compute covariance matrix from data points x_i
 2. Perform eigenanalysis to find principal components
 3. Create orthonormal matrix U from eigenvectors
 4. Transform data: $z = U^T x$
- Result: diagonalized covariance matrix with independent components

Dimensionality Reduction Applications

- Basis vector interpretation
 - PCA creates basis images capturing frequency components
 - Higher-order components represent higher frequency variations
 - Can threshold eigenvalues to keep only significant components
- Practical benefits for image processing
 - 512×512 image = 2^{18} dimensional vector (computationally intensive)
 - PCA reduces to manageable lower dimensions
 - Removes noise while preserving important signal information
- Energy optimization property
 - PCA captures maximum energy given fixed number of coefficients
 - Most optimal among all unitary transforms (vs Fourier, DCT)

Neural Networks vs Traditional Features

- Philosophical difference in approach
 - Traditional: handcraft features → train classifier separately
 - Neural networks: integrate feature extraction with task optimization
 - Networks learn features that maximize performance for specific task
- When to use traditional approaches:
 - Limited training data availability
 - Lack of labeled data (expensive to obtain)
 - Safety-critical applications
- Neural network advantages
 - Trainable end-to-end pipeline
 - Task-specific feature optimization
 - Can find complex nonlinear representations in high-dimensional spaces

Chat with meeting transcript: <https://notes.granola.ai/t/5783a6cd-60bc-4b79-b902-cda21c3fe590>

Logistics & Course Admin

- Partner selection for project: notify TA (PNO) ASAP — ideally this week
- Project spans 6–8 weeks; five broad topic areas to choose from:
 - Image representation
 - Grouping and segmentation
 - Tracking
 - Recognition
 - Reconstruction
- If you have a topic preference (e.g. Gaussian splats), declare it now — otherwise one gets assigned
- Assignment 1: ~5 problems, uploading today, ~3 weeks to complete
 - Includes problems on phase transfer and downsampling/filtering implementation
- Three assignments total, each ~3 weeks
- Final exam: ~20–30 multiple choice questions on fundamentals (no math)
- Paper presentations begin late this week or next week

Sampling & Aliasing

- Sampling = multiplying a signal by an impulse train to pick up discrete values
- Fourier transform of a sampled signal = convolution of original FT with FT of impulse train
 - Key result: sampling preserves frequency-domain properties, not just spatial/time-domain values
- Nyquist rate: must sample at $\geq 2\times$ the maximum frequency in the signal
 - Under-sampling $\rightarrow$ aliasing (e.g. chariot wheel appearing to spin backwards in old films)
 - Over-sampling $\rightarrow$ wastes storage, compute, and hardware clock cycles; also risks sampling noise rather than signal
- Always low-pass filter before downsampling
 - Restricts maximum frequency $\rightarrow$ allows lower valid sampling rate
 - Prevents aliasing artifacts visible in downsampled images
- Preferred filter: Gaussian (not box/gate function)
 - Gate function has sharp edges $\rightarrow$ high-frequency ringing in frequency domain (sinc response)
 - Gaussian convolved with Gaussian = Gaussian; smooth falloff, no ringing
 - Gate functions are also impractical to implement in hardware

Image Pyramids & Wavelets

- Image pyramid: iteratively filter then downsample (e.g. 512 $\rightarrow$ 256 $\rightarrow$ 128...)
 - Each level loses high-frequency detail (the “residue”)
 - Residue = original image minus upsampled approximation $\rightarrow$ looks like edges (high-frequency content)
- Wavelets extend this: multi-resolution analysis at varying window sizes
 - Short-term Fourier transform: fixed-size time chunks $\rightarrow$ fixed frequency resolution
 - Wavelets: variable window size $\rightarrow$ finer time-frequency tiling
 - Extra computation, but enables analysis impossible with standard Fourier transform
- Applications: image compression (lossy), turbulent flow analysis, U-Net-style architectures
- Wavelet/multi-resolution ideas have faded from core research but persist in architectures like U-Net

Principal Components Analysis (intro — to be continued Wednesday)

- PCA = finding directions of maximum variance in data
- Connects to frequency analysis: most natural signals/images are low-frequency (smooth regions dominate)
- Unitary transforms: inverse = transpose (e.g. Fourier, DCT, PCA basis)
- Covariance matrix is the key tool for measuring variance in multiple dimensions
 - For zero-mean vector $\mathbf{x}$: $\mathbf{C}(\mathbf{x}) = \mathbb{E}[\mathbf{x} \mathbf{x}^T]$
 - Diagonal elements = variances of individual components
 - Off-diagonal = covariances (related to correlation coefficient ρ)
- PCA reduces to eigen-analysis of the covariance matrix — covered Wednesday

Next Steps

- Vedant
 - Submit project topic preference and partner to TA this week
 - Review covariance matrix structure before Wednesday's class (especially if unfamiliar with PCA)
 - Complete Assignment 1 once uploaded (due ~3 weeks from today)

Chat with meeting transcript: <https://notes.granola.ai/t/93371afd-637a-40df-b6b1-4b86da7b6bee>

Spatial Frequency & Fourier Transform

- Fourier transform = projecting a signal onto basis functions (sine/cosine waves) and measuring the importance of each
- In 2D (images), spatial frequency is a **vector** ($\mathbf{u}, \mathbf{v}$) — not a scalar

- Has both magnitude (rate of intensity change) and direction (orientation of change)
 - Vertical edges carry different spatial frequencies than horizontal edges
- Low spatial frequency → slow intensity change across image (e.g., plain wall)
- High spatial frequency → rapid intensity change (e.g., edges, fine textures)
- Natural images are predominantly low frequency — intensity doesn't change much across most regions

Magnitude vs. Phase

- Fourier transform produces a complex output: magnitude + phase
 - Magnitude = which frequencies are present (intuitive, easy to interpret)
 - Phase = where zero-crossings occur; captures spatial organization of the signal
- Phase carries more information than magnitude
 - Demo: reconstructing with zebra phase + cheetah magnitude → looks like a zebra
 - Reconstructing with cheetah phase + zebra magnitude → looks like a cheetah
- Application: style transfer (day → night) works by swapping phase from nighttime images
- Encoding note: Fourier transform doubles data (real → complex), so compression uses Discrete Cosine Transform (DCT) instead — outputs real numbers only

Sampling & Nyquist Criterion

- To reconstruct a signal perfectly, must sample at $\geq 2\times$ the highest frequency present
 - This is the **Nyquist sampling criterion**
- Under-sampling → aliasing (cannot reconstruct original signal)
- Over-sampling → reconstruction works, but wastes computation and storage
- Critical sampling = exactly $2\times$ max frequency; minimum needed for perfect reconstruction
- Same principle applies in 2D (images): sampling rate must account for spatial frequency in both dimensions
- Next class: math behind aliasing and 2D sampling

Chat with meeting transcript: <https://notes.granola.ai/t/1adbcfac-a5f9-482b-bfe1-cb4432ae1c62>